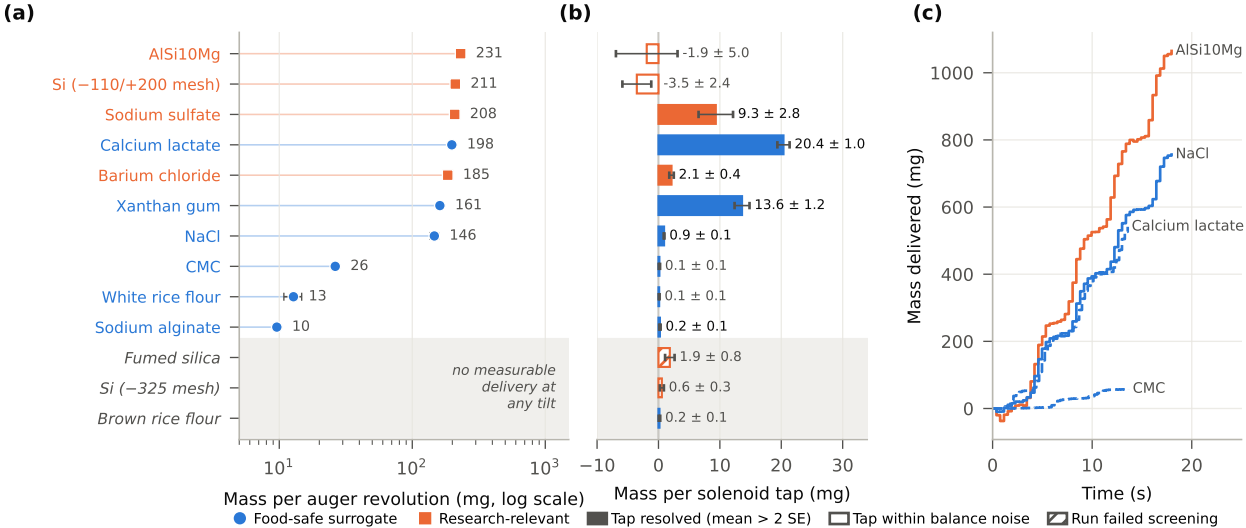
(a)
AlSi10Mg
Si (−110/+200 mesh)
Sodium sulfate
Calcium lactate
Barium chloride
Xanthan gum
NaCl
CMC
White rice flour
Sodium alginate
Fumed silica
Si (−325 mesh)
Brown rice flour
231
211
208
198
185
161
146
26
13
10
no measurable delivery at any tilt
10^1
10^2
10^3
Mass per auger revolution (mg, log scale)
(b)
-1.9 ± 5.0
-3.5 ± 2.4
9.3 ± 2.8
20.4 ± 1.0
2.1 ± 0.4
13.6 ± 1.2
0.9 ± 0.1
0.1 ± 0.1
0.1 ± 0.1
0.2 ± 0.1
1.9 ± 0.8
0.6 ± 0.3
0.2 ± 0.1
−10
0
10
20
30
Mass per solenoid tap (mg)
(c)
AlSi10Mg
NaCl
Calcium lactate
CMC
Mass delivered (mg)
1000
800
600
400
200
0
0
10
20
Time (s)
Food-safe surrogate
Research-relevant
Tap resolved (mean > 2 SE)
Tap within balance noise
Run failed screening